NCFE Level 2 Functional Skills Qualification in Mathematics (603/5060/X)

Paper number:	Practice P001269
Section B:	Calculator Test



Time allowed: 1 hour 30 minutes

Learner instructions

- Answer **all** questions.
- Read each question carefully.
- Write your answers in the spaces provided.
- Show your working, as marks may be awarded for working.
- State units in your answers, where appropriate.
- Check your work.
- Use π = 3.14

Learner information

- Section B contains **Activities 2, 3** and **4**.
- The maximum mark for this section is **45**.
- The marks available for **each** question are shown in brackets.

Resources

You will need a:

- pen, with black or blue ink
- pencil and eraser
- 30 cm ruler
- protractor
- calculator.

If extra pages are used, please make sure your name and centre name are on them and they are securely fastened to this booklet.

Please complete the details below clearly and in BLOCK CAPITALS.

Learner name _____

Centre name _____

Learner number | | Centre number | |

Do not turn over until the invigilator tells you to do so.

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Activity 2: Films

2 (a) Liam downloads films from the internet.

The table shows the number of films he downloaded last month.

	Action	Comedy	Sci-Fi	Total
Less than 1 hour long	5	2	4	11
Exactly 1 hour long	2	4	6	12
More than 1 hour long	1	1	3	5
Total	8	7	13	28

He chooses one of these films at random.

What is the probability that the film he chooses is an action film that is at least 1 hour long?

[2 marks]

Your answer:

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2 (b) Liam thinks he might watch either Rainbow Prism or The Infinity Zone.

He looks at the scores given by different newspapers.
The scores are out of 100

	Rainbow Prism	The Infinity Zone
Daily Times	75	75
London Post	63	68
City Observer	52	not scored
Newcastle Daily	38	50
Entertainment	70	74
Britain Today	83	65
This Week	60	79
Northern Gazette	41	40
Sunday Extra	52	63

Which of the two films would you suggest Liam watches next?
Use **two** appropriate statistical measures to explain your answer.

[3 marks]

Your answer:

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2 (c) Screen images are made up of units of light called pixels.

Pixels per square inch (PPI^2) is a measure of image quality.
The higher the pixels per square inch, the better the image quality.

$$PPI^2 = \frac{N}{A}$$

Where: N is the number of pixels
A is the area of the screen in square inches

Liam's old screen has a PPI^2 value of 9800
He wants to buy a new screen with a better image quality.

Liam sees a rectangular screen that has:

- a width of 101.6 cm
- width : height ratio of 16 : 9
- 8.28 million pixels

He works out that this will have a better image quality than his old screen.

Is Liam correct? Show your working.

Use the conversion: 1 inch = 2.54 cm

[4 marks]

Your answer:

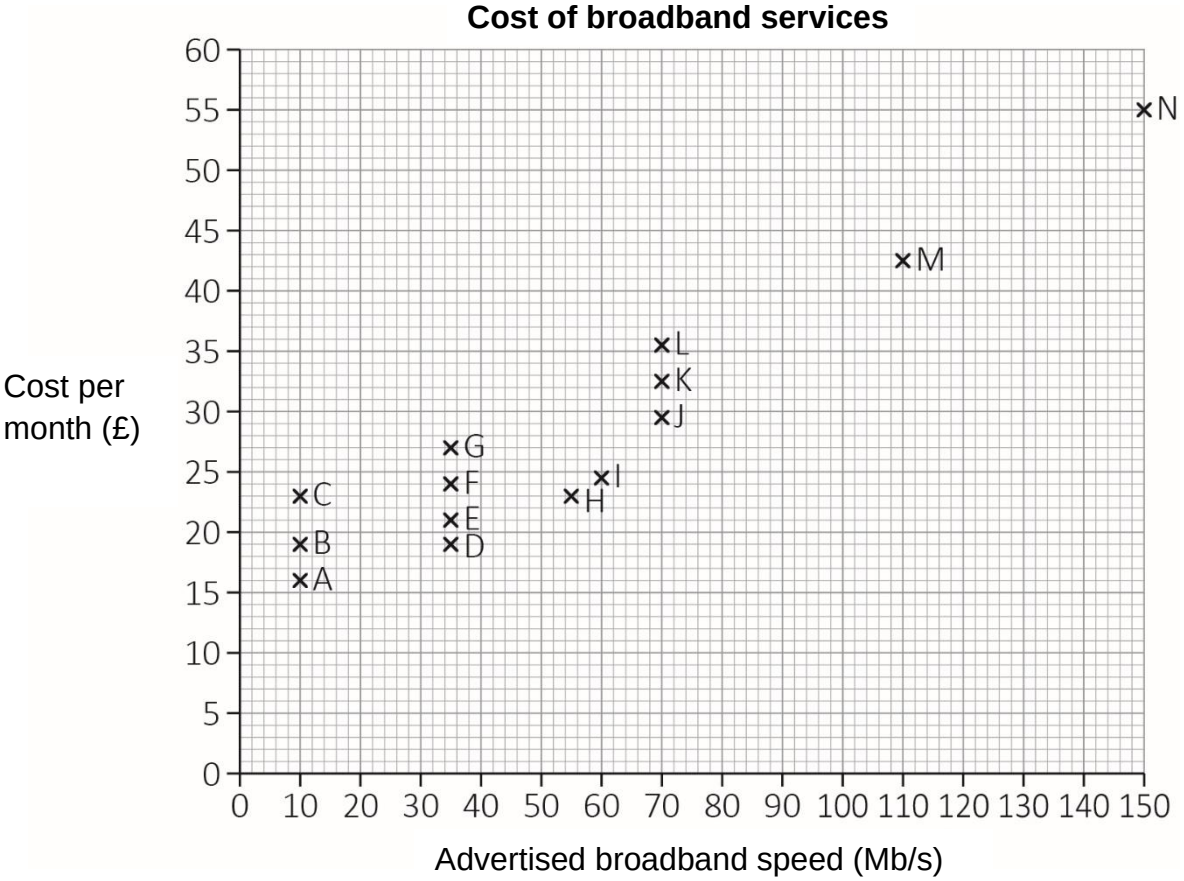
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2 (d) Liam searches online for a choice of broadband services.

He finds this graph:



Broadband speed is measured in Mb/s (Megabits per second).

The lettered crosses in the graph represent the different services.

What is the modal speed of these broadband services?

[1 mark]

Your answer: **Mb/s**

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2 (e) Using the graph from **2 (d)**, what percentage of these broadband services cost over £25 per month?

Give your answer to 2 decimal places.

[2 marks]

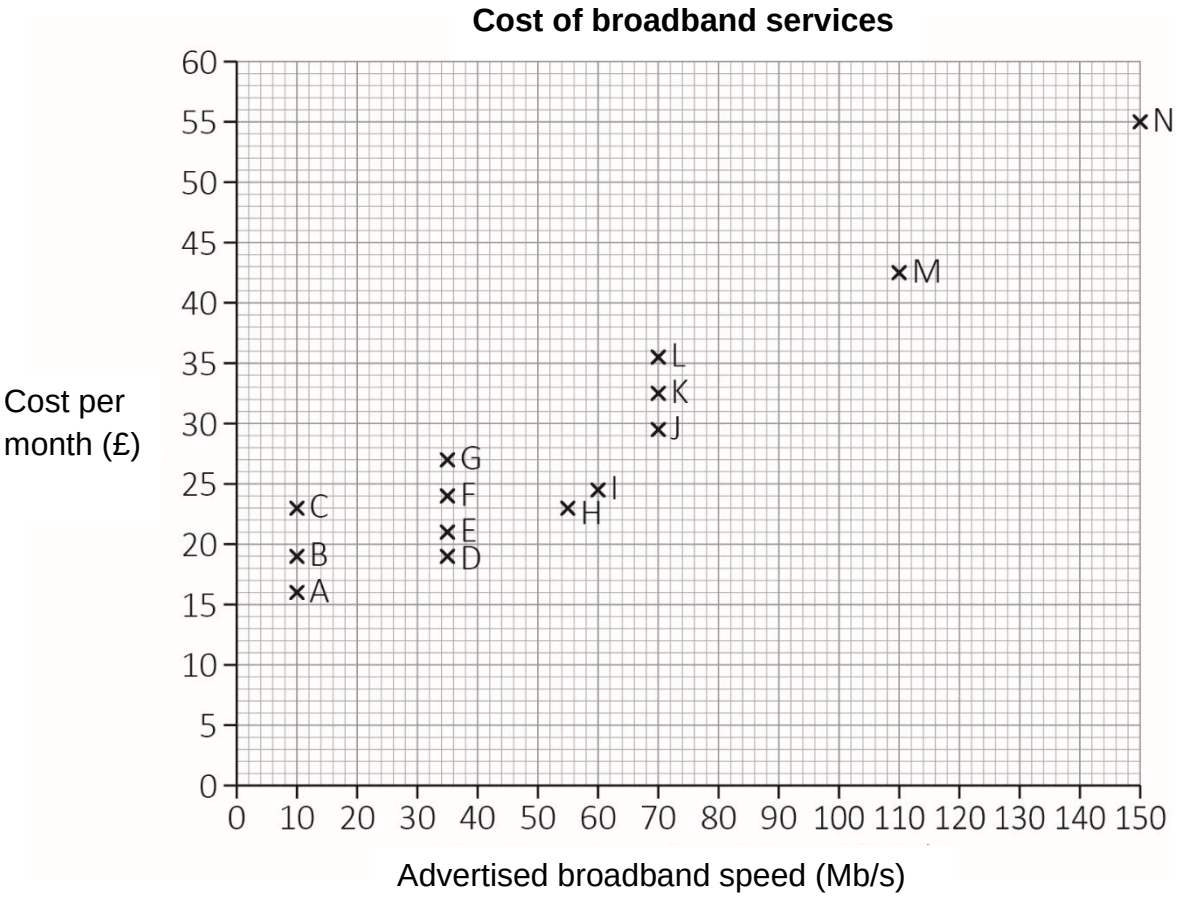
Your answer: **%**

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2 (f) This is the graph from **2 (d)**.



The actual speed of a broadband service depends on how far the customer's house is from the nearest connection box.

Distance from house to connection box (to the nearest m)	Percentage of advertised speed
0-100 m	100%
101-500 m	85%
501-1000 m	70%

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Liam's house is 350 m from the connection box.

Liam wants to be able to download a 7000 Mb film in less than 2 minutes.

What is the cheapest broadband service he can choose?
Show your working.

[3 marks]

Your answer:

[Total marks: 15]

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Activity 3: Building a pond

3 (a) Aria is involved in a community project group.

The group are planning to build a pond in the garden of the local hospital.
The pond has a path around it.

On a scale plan, the width of the path is 6.7 cm

The scale used on the plan is 1 : 25

Calculate the actual width of the path.

[1 mark]

Your answer: **cm**

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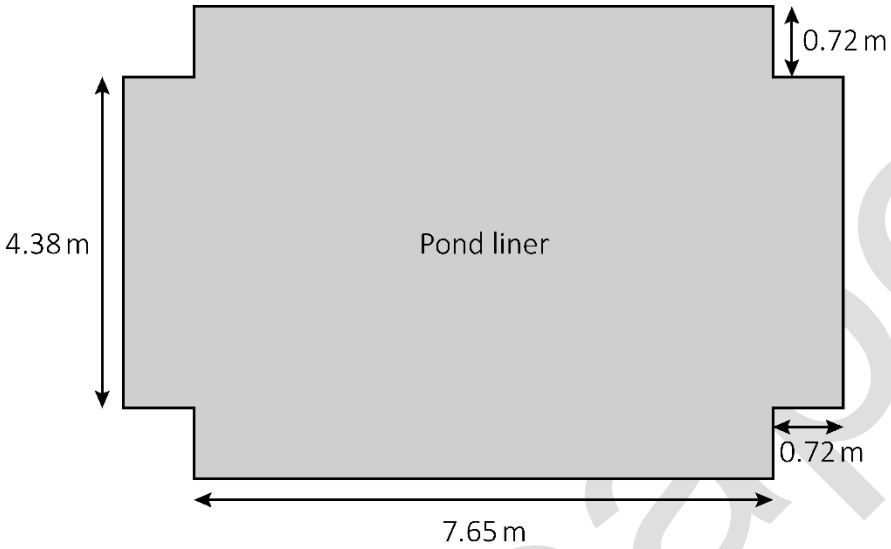
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3 (b) The pond will be a cuboid. It will be lined with a sheet of plastic pond liner. The pond liner covers the bottom and sides of the pond.

The pond liner needs to be cut to this shape to exactly fit in to the pond without any overlaps.

The dimensions of the pond liner are shown in this diagram:



Sheets of pond liner are available in these sizes:

Pond liner	Width	Length	Price
A	4.85 m	8.3 m	£76
B	5.45 m	8.7 m	£96
C	5.85 m	9.1 m	£130
D	6.25 m	9.5 m	£151
E	6.65 m	9.9 m	£172

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Aria needs to buy the cheapest pond liner that is big enough to fit the pond.

Which pond liner should she buy?
Show your working.

[3 marks]

Your answer:

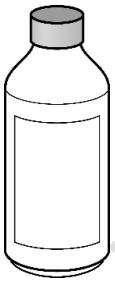
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3 (c) The group ask Aria to buy chemical treatment for the pond water to make it suitable for fish.

One bottle contains enough chemical to treat 1300 gallons of water.



1 m^3 = 220 gallons

Aria adds water to the pond until it is 92% full.

How many bottles of chemical treatment will she need?
Show your working.

[4 marks]

Your answer: **bottles**

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3 (d) Aria needs to work out how many fish can be put in the pond.
She wants to have both Koi fish and Goldfish.

The number of fish is based on the **surface area** of the water in the pond.

Koi fish need 1.2 m^2 each.
Goldfish need 0.6 m^2 each.

Aria wants to have a mixture of Koi fish and Goldfish in the ratio 1 : 4

What is the maximum number of Koi fish **and** Goldfish that Aria can put in the pond, if she keeps the ratio 1 : 4?

[3 marks]

Your answer:

Koi fish

Goldfish

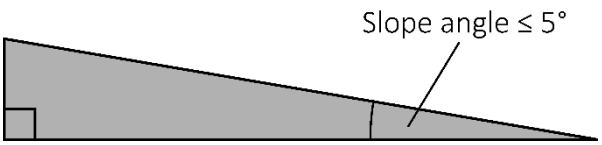
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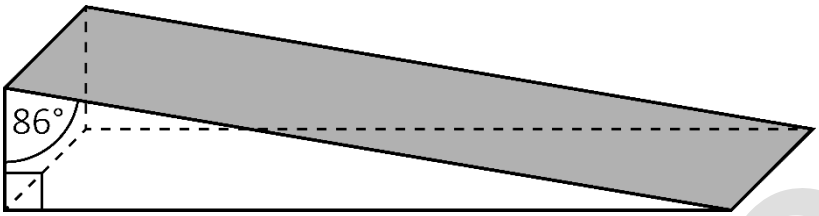
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3 (e) The group need to build a ramp between the hospital door and the gardens.

The angle of the slope of the ramp must not be more than 5°



This is a diagram of the ramp:



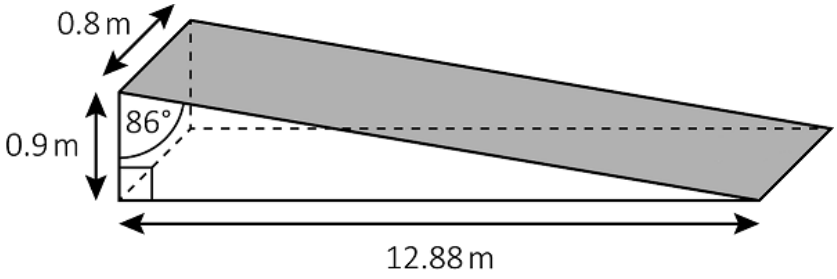
Calculate the angle of the slope.

[1 mark]

Your answer: °

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3 (f) This diagram shows the measurements of the ramp:



Draw the plan view of the ramp as seen from above.
Include the dimensions on your drawing.

[1 mark]

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3 (g) The group need to order enough concrete to make the ramp.

The formula to find the volume of a triangular prism is: $\frac{L \times H \times W}{2}$

Where:

L is length
H is vertical height
W is width

Calculate the volume of concrete that is needed to build the ramp.

[2 marks]

Your answer: **m^3**

[Total marks: 15]

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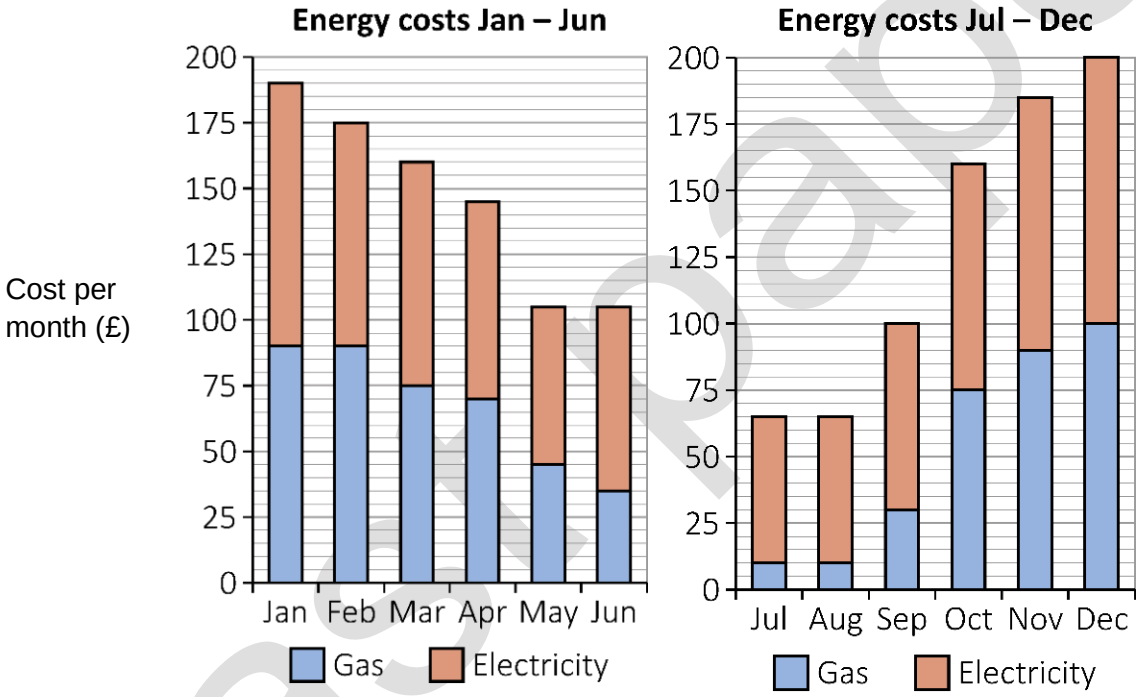
Activity 4: Saving energy costs

4 (a) Eli needs to complete a project on energy costs for his college course.

Eli's family currently use Xpower for gas and electricity.
They pay monthly by direct debit.

Xpower uses the **median** monthly cost for January to June to set the amount of their monthly direct debit payments for July to December.

The graphs show their energy costs for last year.



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Examiner use only

Eli works out that they have paid too much between July and December.

Calculate the percentage by which they have overpaid.

[4 marks]

Your answer: **%**

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Examiner use only

4 (b) This is Eli's electricity bill for the first quarter of this year:

Total electricity used	£211.90
Dual fuel discount	–£14.90
VAT	£9.85
Total including VAT	£206.85

VAT is calculated after discount.

What percentage has been used to calculate the VAT?

[3 marks]

Your answer: **%**

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Examiner use only

4 (c) Xpower offer a Loyalty Plan.
The formula Xpower use to work out the cost for the Loyalty Plan is:

$$(D \times 365) + (P \times G)$$

Where:

D is daily charge (in £)
P is price per kWh (in £)
G is gas used per year (in kWh)

For this plan the charges are:

- Daily charge: 32p
- Price per kWh: 3.10p

Last year Eli's family used 17563kWh of gas.
This cost £719.38

Should Eli's family choose the Loyalty Plan for their gas?
Show your working.

[2 marks]

Your answer:

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Examiner use only

4 (d) The average UK gas bill is 3.875 times more than the average UK electricity bill.

[2 marks]

(i) Write 3.875 as an improper fraction.

(ii) Write 3.875 as a percentage.

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Examiner use only

4 (e) Eli looks for information on the cost of leaving gadgets on standby when they are not being used.

Eli's gadgets are a TV, computer, printer, phone charger and two games consoles.

He finds several reports stating how much electricity a gadget uses when left on standby, but they do not agree.

Annual saving per gadget	Number of reports
£0 - £4.99	1
£5 - £9.99	4
£10 - £14.99	5
£15 - £19.99	2

Use an estimate of the mean to find how much money Eli could save if he switched off his gadgets, instead of leaving them on standby.

[4 marks]

Your answer: **£**

[Total marks: 15]

This is the end of the assessment.

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